Household's Utility Function

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Utility from services $c(t)$:

$$\boxed{\frac{\varepsilon}{\varepsilon - 1} \cdot c(t)^{\frac{\varepsilon-1}{\varepsilon}}} \qquad w/ \quad \varepsilon > 1$$

Utility from real wealth.

- Real wealth: real stock of government bonds

  $\downarrow$ $w(t)$

- Average real wealth: $\bar{w}(t)$
- Relative real wealth: $[w(t) - \bar{w}(t)]$

  $\downarrow$ enters utility function

- $\sigma: \mathbb{R} \to \mathbb{R}$, strictly increasing, strictly concave

$$\boxed{\sigma(w(t) - \bar{w}(t))}$$

Aggregate utility.

$\delta > 0$ time discount rate

services

relative real wealth

$$\int_0^{\infty} e^{-\delta t} \left[\frac{\varepsilon}{\varepsilon - 1} c(t)^{\frac{\varepsilon-1}{\varepsilon}} + \sigma\big(w(t) - \bar{w}(t)\big) \right] dt$$